JURNAL PRAKTIKUM

High Frequency

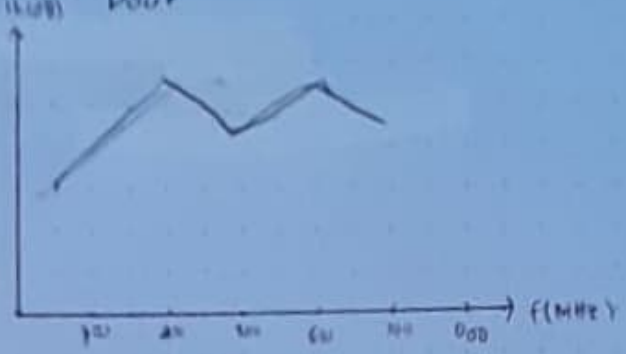
Kelompok : 3
Kelas : 3NK

Anggota :
- Bima Rizki F
- Dicky Nur A
- Dwi Agustina
- Dyekie Wijaya

Vin = 400 mV

A. Insertion Loss => IL = 20 log $\frac{V_{in}}{V_{out}}$

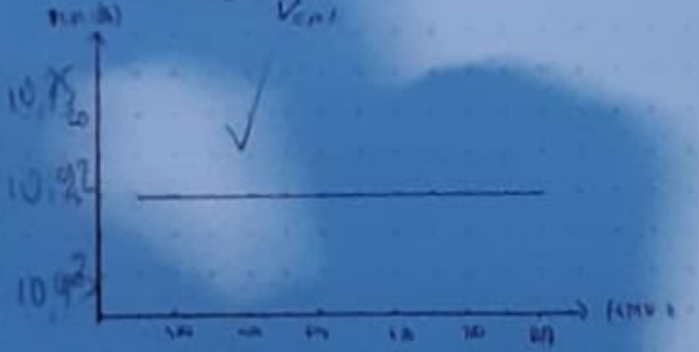
f (MHz)	Vin	Vout	IL
100	400 mV	200 mV	6,0
200		220 mV	5,2
300		300 mV	2,5
400		295 mV	2,6
500		230 mV	2,8
600		220 mV	2,9



- f = 300 MHz
  IL = 20 log $\frac{400}{200}$ =
- f = 400 MHz
  IL = 20 log $\frac{400}{220}$ =
- f = 500 MHz
  IL = 20 log $\frac{400}{}$ =
- f = 600 MHz
  IL = 20 log $\frac{400}{}$ =
- f = 700 MHz
  IL = 20 log $\frac{400}{}$ =
- f = 800 MHz
  IL = 20 log $\frac{400}{}$ =

B. Coupling Factor => CF = 20 log $\frac{V_{in}}{V_{CPL}}$

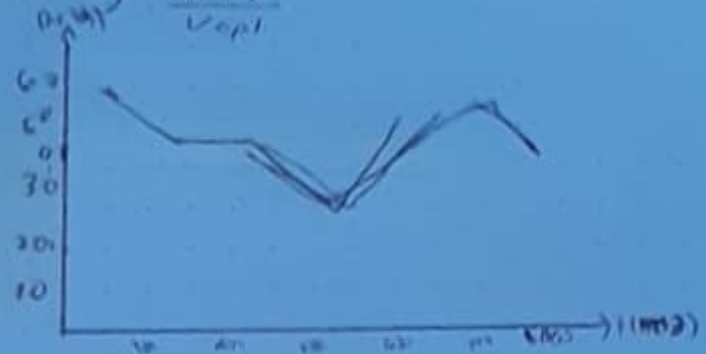
f (MHz)	Vin	VCPL	CF
300	400 mV	70 mV	15,1
400		71 mV	15,0
500		70 mV	15,1
600		70 mV	15,1
700		70 mV	15,1
800		69,5 mV	15,1



- f = 300 MHz
  CF = 20 log $\frac{400}{70}$ = 15,1
- f = 400 MHz
  CF = 20 log $\frac{400}{71}$ = 15,0
- f = 500 MHz
  CF = 20 log $\frac{400}{70}$ = 15,1
- f = 600 MHz
  CF = 20 log $\frac{400}{70}$ = 15,1
- f = 700 MHz
  CF = 20 log $\frac{400}{70}$ = 15,1
- f = 800 MHz
  CF = 20 log $\frac{400}{69,5}$ = 15,2

* Directivity $\Rightarrow$ Dir = 20 log $\frac{V_{ref}}{V_{opl}}$

f (MHz)	V_{opl} (mV)	V_{ref} (mV)	Dir (dB)
300	70	0.2	
400	73	0.3	
500	70	0.3	
600	70	0.5	
700	70	0.2	
800	69.5	0.3	



• f = 300 MHz
Dir = 20 log $\frac{0.2}{70}$ =

• f = 400 MHz
Dir = 20 log $\frac{0.3}{73}$ =

• f = 500 MHz
Dir = 20 log $\frac{0.3}{70}$ =

• f = 600 MHz
Dir = 20 log $\frac{0.5}{70}$ =

• f = 700 MHz
Dir = 20 log $\frac{0.2}{70}$ =

• f = 800 MHz
Dir = 20 log $\frac{0.3}{69.5}$ =